

Walter Thomas Moya Araya

Junior Data Scientist | Junior Data Analyst | Quantitative Analysis

Santiago, Chile | +56 9 3366 9343 | cg.walter.ma@gmail.com

waltermoya.vercel.app | github.com/walterm2482 | linkedin.com/in/walter-moya-araya-a211b9307

Professional Summary

Industrial Engineer with an MSc in Engineering Sciences. Experience in ETL and data quality at PwC Chile and remote quantitative development at ClickAlgo, building reproducible pipelines, evaluating models and developing analytical tools with Python, SQL and statistics.

Technical Skills

Data and BI: Python (Pandas, NumPy), SQL, R, Excel, Power BI, ETL and exploratory data analysis.

Machine learning: scikit-learn, TensorFlow/Keras, clustering, PCA, time series and temporal validation.

Cloud and tools: Google Cloud (BigQuery, Data Fusion, Cloud Storage), C#/.NET, Git, Docker, MLflow.

Selected Projects

K-Means Indicator v2 | Applied Machine Learning

ClickAlgo · 2025

Developed a market regime indicator using K-means clustering, PCA and price, volatility and momentum features. Refactored the code and added visualizations and Telegram alerts.

FSG Ultimate / Smart Grid Ultimate v2.1 | Automation and Risk

cTrader Store · 2024–2026

Developed and published a mean-reversion trading robot in C#/.NET with closed-bar execution, ATR-based targets and three layers of risk controls to manage exposure and losses.

Smart Portfolio Architect | Portfolio Diversification

ClickAlgo · 2024

Automated minimum-correlation asset selection in C# for cTrader. The tool analyzes historical data and compares asset combinations to support portfolio diversification.

Portfolio Optimizer | Python, Dash and Backtesting

GitHub · 2025

Built a Dash application with four allocation methods and rebalancing backtests. Added return, drawdown and correlation charts for strategy comparisons.

Professional Experience

Quantitative & AI Developer | Independent Project

10/2024 – Present

- Built a Python research platform for market data, feature engineering, predictive models and backtesting; automated data quality checks and result tracking.
- Applied temporal validation and out-of-sample evaluation to reduce information leakage between model training and evaluation on financial time series.

Junior Programmer | ClickAlgo, United Kingdom · Remote

11/2023 – 10/2024

- Developed quantitative indicators in C#/.NET using clustering, Kalman filters and volatility analysis; delivered tested, documented modules and provided technical support to international users.
- Published a technical article on minimum-correlation portfolios using Pearson, Spearman and Kendall.

Analyst Trainee | PwC Chile

08/2022 – 11/2022

- Optimized historical inventory loads of hundreds of thousands of records to **under 30 seconds per file**.
- Supported ETL pipelines on Google Cloud using Python and REST APIs, schema validation and data quality checks for Power BI dashboards.

Education

MSc in Engineering Sciences, Industrial Engineering

08/2021 – 06/2023

Universidad Diego Portales · **Highest distinction** · Thesis grade: 6.8/7.0.

Thesis: construction machinery detection and tracking with YOLOv8 and BoT-SORT.

Industrial Engineering Degree

03/2016 – 07/2023

Universidad Diego Portales · Graduated with distinction.

Selected Certifications

MLOps Specialization — Duke University (2026) · Machine Learning — DeepLearning.AI / Stanford (2025).

IBM Data Science Professional Certificate (2025) · Google Data Analytics Professional Certificate (2025).